

HDOC DAY 35

PROBLEM

Write a menu-driven C program to read a variable-length square integer array and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using a diagonal traversal. Begin with the element in the top-left corner and traverse successive diagonals alternately in the upward-right and downward-left directions.
4. Exit.

Example: For the following input matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

1. Main diagonal elements: 1 5 9. All the main diagonal elements are distinct.
2. Sum of main diagonal elements: 15
3. Diagonal traversal: 1 2 4 7 5 3 6 8 9

C CODE

```
#include <stdio.h>

void issame()

{

    int n,m;

    printf("enter the size of 2d array or matrix(row*column)\n");

    scanf("%d%d",&n,&m);
```

```
int arr[n][m];
for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        scanf("%d",&arr[i][j]);
    }
}

for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        printf("%d ",arr[i][j]);
    }
    printf("\n");
}

printf("\nthe main diagonol elements are:\n");
for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        if(i==j)
        {
            printf("%d\t",arr[i][j]);
        }
    }
}

int count=0;
```

```

for(int i=0;i<n-1;i++)
{
    for(int j=0;j<m-1;j++)
    {
        if(arr[i][j]==arr[i+1][j+1])
            count=0;
        else
            count++;
    }
}
if(count==0)
    printf(",not all elements are different");
else
    printf(",all elements are different");
}

void sum()
{
    int n,m;
    printf("enter the size of 2d array or matrix(row*column)\n");
    scanf("%d%d",&n,&m);
    int arr[n][m];
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<m;j++)
        {
            scanf("%d",&arr[i][j]);
        }
    }
}

```

```

        for(int i=0;i<n;i++)
        {
            for(int j=0;j<m;j++)
            {
                printf("%d ",arr[i][j]);
            }
            printf("\n");
        }
int sum=0;
printf("\nthe main diagonol elements are:\n");
for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        if(i==j)
        {
            printf("%d\t",arr[i][j]);
            sum=sum+arr[i][j];
        }
    }
}
printf("\nthe sum of main diagonls is%d\n",sum);
}
void traversal()
{
    int n,m;
    printf("enter the size of 2d array or matrix(row*column)\n");
    scanf("%d%d",&n,&m);

```

```

int arr[n][m];
for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        scanf("%d",&arr[i][j]);
    }
}

for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        printf("%d ",arr[i][j]);
    }
    printf("\n");
}

for(int i=0;i<(n*2-1);i++)
{
    if(i%2==0)
    {
        for(int a=i;a>=0;a--)
        {
            for(int b=0;b<=i;b++)
            {
                if(a+b==i && a<n && b<n)
                {
                    printf("%d\t",arr[a][b]);
                }
            }
        }
    }
}

```

```

        }
    }
}
else if(i%2!=0)
{
    for(int a=0;a<=i;a++)
    {
        for(int b=i;b>=0;b--)
        {
            if(a+b==i && a<n && b<n)
            {
                printf("%d\t",arr[a][b]);
            }
        }
    }
}
}

int main()
{
    int ch;

    printf("choice 1 Determine whether all the elements on its main diagonal are
distinct.\n");

    printf("choice 2 Find the sum of its main diagonal elements.\n");

    printf("choice 3 isplay its elements using a diagonal traversal. Begin with the element
in the top-left corner and traverse successive diagonals alternately in the upward-right
and downward-left directions.\n");

    printf("exit is choice 4\n");

    printf("enter your choice\n");

    scanf("%d",&ch);

```

```
switch (ch)
{
    case 1:
    {
        issame();
        break;
    }
    case 2:
    {
        sum();
        break;
    }
    case 3:
    {
        traversal();
        return 0;
    }
    case 4:
    {
        printf("exit\n");
        return 0;
    }
    default:
    {
        printf("invalid choice\n");
        break;
    }
}
```

```
return 0;
```

```
}
```